Risk-Averse Humanitarian Logistics

Topic Pitch

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Motivation & Focus

Motivation:
Standard humanitarian logistics models optimize for the "average" disaster. However, in catastrophic events, average planning fails, leading to massive resource shortages.

Research Question:
How does shifting from a risk-neutral to a risk-averse $(CVaR_\alpha)$ optimization approach impact unmet demand and response times in a multi-period disaster relief network?

Intended Contribution:
We will extend classical disaster relief models to include multi-period networks with stochastic transport times, comparing CVaR policies against standard risk-neutral policies through computational experiments.

Risks & Proposed Modifications

Most Difficult Parts:

- Implementing the CVaR objective function in code (translating the Rockafellar & Uryasev LP formulation into Python).
- Sourcing or synthetically generating a realistic dataset for extreme disaster scenarios.

Proposed Modifications:

The original topic suggests potentially looking at "distributionally robust models." To keep the scope manageable and ensure a high-quality computational implementation, we propose restricting our focus strictly to CVaR/Mean-CVaR, dropping the distributionally robust component.